

Architectural floor plan of the 11th floor of the World Trade Center, showing mechanical, electrical, and plumbing (MEP) systems. The plan includes various rooms such as offices, meeting rooms, restrooms, and a central core. MEP equipment like air handling units (AHUs), fans, and ductwork are labeled with codes like G (gas), FFB (fire-fighting), and C (cooling). Dimensions and elevations are provided for many components. A note in the upper right corner states: "DUCT WORK TO BE ROUTED THROUGH TWO (2) EXISTING BEAM CUTS. 30x10 DUCT SHALL BE ALIGNED WITH BEAM RUNNING PLAN NORTHEAST TO CORNER OF STRUCTURE." The plan is oriented with Seventh Avenue to the left and Broadway to the right.

- 1 DUCTWORK TO BE CONSTRUCTED OF RECTANGULAR 16-GAUGE OUTER DUCT WITH/ ACoustICAL LINING, TIED LINER AND STANDARD GAUGE PERFORATED INNER DUCT TO MATCH BUILDING STANDARD FOR FAN DISCHARGE DUCTWORK.
- 2 EXPANDED DUCTWORK SHALL BE OVAL OR ROUND SPIRAL DUCT CONSTRUCTED OF GALVANIZED STEEL, WITH ACoustICAL LINING AS PER SPECIFICATIONS, FABRICATE AND WELD PER SMAW OR FCAW PROCESSES WITH ALL SEAMS AIRTIGHT AND CONNECTIONS SUITABLE FOR PRESSURE CLASS, SUPPORT AND HANG DUCT PER CODE REQUIREMENTS.
- 3 UNIT LOCATION TO BE COORDINATED WITH OFF LAYOUT, LOCATION AND POSITIONING SHALL BE ALLOWED FOR ALL CLEARANCES AS SPECIFIED IN CONTRACT DOCUMENTS. IF HUNG, UNIT SHALL BE VIBRATION INSULATED AS DESCRIBED FOR MECHANICAL EQUIPMENT IN SPECIFICATIONS.

1. ALL WORK SHALL CONFORM TO THE LATEST BUILDING STANDARDS. IF IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO OBTAIN A COPY OF THE BUILDING STANDARDS, MEET WITH THE PROJECT ENGINEER IN ORDER TO BECOME TOTALLY FAMILIAR WITH THE BUILDING CONSTRUCTION RULES. THERE SHALL BE NO DEVIATION FROM THE BUILDING STANDARDS WITHOUT PRIOR WRITTEN APPROVAL FROM THE BUILDING ENGINEER.

2. WHEN MECHANICAL WORK IS SUBCONTRACTED IT SHALL BE THE MECHANICAL CONTRACTORS RESPONSIBILITY TO COORDINATE SUBCONTRACTORS AND THE ASSOCIATED CONTRACTS, WHEN DISCREPANCIES ARISE PERTAINING TO WHICH CONTRACTOR PROVIDES A PARTICULAR ITEM OF THE MECHANICAL CONTRACT OR WHICH CONTRACTOR PROVIDES FINISHING MANAGEMENT IN ORDER TO BECOME TOTALLY FAMILIAR WITH THE BUILDING CONSTRUCTION RULES. THERE SHALL BE NO DEVIATION FROM THE BUILDING STANDARDS WITHOUT PRIOR WRITTEN APPROVAL FROM THE MECHANICAL CONTRACTOR. IT SHALL BE BROUGHT TO THE ATTENTION OF THE MECHANICAL CONTRACTOR, WHOSE DECISION SHALL BE FINAL.

3. ALL MISCELLANEOUS STEEL REQUIRED TO ENSURE PROPER INSTALLATION AND AS SHOWN ON THE DRAWINGS FOR THE LIGHTING FIXTURES SHALL BE FURNISHED AND INSTALLED BY THE MECHANICAL CONTRACTOR AND COORDINATED WITH THE GENERAL CONTRACTOR FOR COORDINATION AMONG ALL TRADES.

4. PROVIDE ALL MATERIALS, EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

5. INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH ALL APPLICABLE CODES, RECOMMENDATIONS, CONTRACT DOCUMENTS, AND APPLICABLE CODES AND REGULATIONS.

6. PROVIDE VIBRATION ISOLATION FOR ALL MECHANICAL EQUIPMENT TO PREVENT TRANSMISSION OF VIBRATION TO BUILDING STRUCTURE.

7. PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS CONNECTED TO AND WITHIN 50 FEET OR 50 PIPE DIAMETERS IF ISOLATED EQUIPMENT REFER TO SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL PIPING, DUCTWORK AND EQUIPMENT VIBRATION ISOLATION.

8. LOCATE AIR TEMPERATURE, PRESSURE AND FLOW MEASURING DEVICES IN ACCESSIBLE AREAS THAT ARE NOT IN THE PATH OF DIRECT DUCT UP- AND DOWNSTREAM AS RECOMMENDED BY THE MANUFACTURER.

9. COORDINATE ALL EQUIPMENT CONNECTIONS WITH MANUFACTURER'S CERTIFIED DRAWINGS, COORDINATE AND PROVIDE ALL DUCT AND PIPING TRANSITIONS REQUIRED FOR THE FINAL EQUIPMENT CONNECTIONS TO FURNISHED EQUIPMENT. FIELD VERIFY COORDINATE ALL DUCT AND PIPING DIMENSIONS BEFORE FABRICATIONS.

10. THE LOCATIONS OF ALL ITEMS SHOWN ON THE DRAWINGS OR CALLED FOR IN THE CONTRACT DOCUMENTS THAT REQUIRE FIELD VERIFICATION SHALL BE APPROXIMATELY 10% THE EXACT LOCATIONS NECESSARY TO SECURE THE BEST CONDITIONS AND RESULTS MUST BE DETERMINED BY THE PROJECT SITE CONDITIONS AND APPROVED BY THE APPROVAL OF THE ENGINEER BEFORE BEING INSTALLED, DO NOT SCALE DRAWINGS.

11. PROVIDE ACCESS PANELS FOR INSTALLATION IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVICE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER CONCEALED MECHANICAL EQUIPMENT. ACCESS PANELS SHALL BE TURNED OVER TO GENERAL CONTRACTOR INSTALLATION.

12. ALL DUCTWORK, PIPING AND EQUIPMENT SUPPORTED FORM STRUCTURAL, STEEL SHALL BE SUBMITTED WITH MECHANICAL CONTRACTOR, ALL ATTACHMENTS TO STEEL BAR JOISTS, TRUSSES, OR JOIST GIRDERS SHALL BE AT PANEL POINTS. PROVIDE BEAM CLAMPS MEETING SMASMA STANDARDS.

13. ALL OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUCIT, ETC. SHALL BE FIRE STOPPED WITH A PRODUCT SIMILAR TO 3M OR APPROVED EQUIVALE AS SPECIFIED IN STRUCTURAL DOCUMENTS.

14. ALL AIR CONDITONING DEVICES SHALL BE PREPARED WITH PIPING FOR EACH AIR HANDLING UNIT. ALL TRAP AND PIPED TO NEAREST DRAIN. SEE DETAILS SHOWN ON THE DRAWINGS OR THE CONTRACT SPECIFICATIONS FOR DEPTH OF AIR CONDITIONING CONDENSATE TRAP.

15. CERTAIN ITEMS SUCH AS RISERS AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DAMPERS, ETC., ARE INDICATED ON THE CONTRACT DOCUMENT DRAWINGS FOR FIELD VERIFICATION. THE LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS. LOCATIONS OF ALL SUCH ITEMS SHALL BE INDICATED ON SHOP DRAWINGS BY THE INSTALLING CONTRACTOR FOR FIELD VERIFICATION APPROVED BY THE PROJECT ENGINEER.

16. EXACT LOCATIONS OF ALL THERMOSTATS SHALL BE AS PER ARCHITECT/ENGINEERS REFER TO ARCHITECTURAL DWGS FOR THERMOSTAT MOUNTING HEIGHT. MECHANICAL CONTRACTOR SHALL SUBMIT THERMOSTAT LOCATIONS FOR REVIEW PRIOR TO INSTALLATION.

17. ALL DUCTWORK DIMENSIONS, AS SHOWN ON THE DRAWINGS, ARE INTERIOR DIMENSIONS AND DO NOT INCLUDE INCREASED DIMENSIONS TO ACCOUNT FOR DUCT LINING THICKNESS. REFER TO ARCHITECTURAL DWGS FOR EXACT REQUIREMENTS OF ALL DUCT LINING.

18. COORDINATE DIFFUSER, REGISTER, AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS, LIGHTING, AND OTHER CEILING ITEMS.

19. LOCATE ALL MECHANICAL EQUIPMENT TO UNOBSTRUCTED ACCESS TO UNIT ACCESS PANELS, CONTROLS AND VALVING.

20. PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK SYSTEMS CONNECTED TO AIR HANDLING UNITS, FANS AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION. FLEXIBLE CONNECTION REQUIREMENT SHALL BE FOR POINT OF CONNECTION TO THE EQUIPMENT UNLESS OTHERWISE INDICATED.

21. ALL EQUIPMENT, DUCTWORK & PIPING SHALL BE INSTALLED AT MAXIMUM HEIGHTS ALLOWABLE OR AS REQUIRED TO CLAMP NEW CEILING. DUCTWORK & PIPING SHALL BE OFFSET AS REQUIRED TO AVOID CONFLICTS EVEN IF NOT SPECIFICALLY INDICATED ON PLAN.

22. ALL EXPOSED SUPPLY DUCTWORK SHALL HAVE FLANGED AND GASKETTED CONNECTIONS. ALL EXPOSED DUCTWORK SHALL BE PROTECTED BY INSULATION WRAP. PROVIDE ACOUSTIC LININGS AS REQUIRED BY SPECIFICATIONS.

23. ALL EXPOSED DUCTWORK SHALL BE FREE OF DEFORMITIES, WRITING, STICKERS OR ANY OTHER MARKINGS. DUCT SURFACES SHALL BE TREATED AS REQUIRED FOR PAINTING. REFER TO ARCHITECTURAL DWGS FOR LOCATIONS OF DUCTWORK TO BE PAINTED.

24. ALL DUCTWORK EXPOSED TO MOISTURE SHALL BE ALUMINUM FOR A MIN. OF 10 FEET.

25. ALL SUPPLY DUCTWORK SHOWN TO HAVE ROUND DIMENSIONS SHALL BE SPINAL ROUND DUCTWORK.

26. MOUNT ALL SUPPLY REGISTERS AT SAME ELEVATION. ELEVATIONS SHALL BE INDICATED ON THE SHEETMETAL SHOP DWGS FOR REVIEW BY ARCHITECT/ENGINEER.

27. FOR ROOMS WITH EQUIPMENT RACKS: PRODUCTION OF SHEETMETAL SHOP DRAWINGS SHALL INCLUDE ALL COORDINATION REQUIRED TO INDICATE EXACT LOCATION OF ALL TECHNICAL EQUIPMENT, EQUIPMENT RACK SYSTEMS, ETC. SO AS TO ACCURATELY REVERSE COORDINATED PLACEMENT OF ALL AIR OUTLETS SERVING CRITICAL AREAS.

28. PROVIDE ACCESS DOORS IN DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, ETC. ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.

29. SMOKE DETECTOR SHALL BE FURNISHED AND WIRDED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR MOUNTING THE SMOKE DETECTOR IN DUCTWORKS AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS.

30. MECHANICAL CONTRACTOR TO TAKE PRE-CONSTRUCTION AIR READINGS AT ALL BASE BUILDING MECHANICAL ROOMS. PROVIDE A COPY OF WORK AREA, IDENTIFIED KEYED PLAN SHOWING READING LOCATIONS AND SUBMIT TO ENGINEER PRIOR TO PRODUCTION OF SHEET METAL SHOP DWGS. REFER TO SPECIFICATIONS FOR MORE INFORMATION.

31. AIR BALANCING WORK SHALL BE WITNESSED BY THE BUILDING'S CHIEF ENGINEER OR HIS REPRESENTATIVE AND A COPY OF THE CERTIFIED BALANCING REPORT SHALL BE SUBMITTED TO THE BUILDING MANAGEMENT. PROVIDE 48 HOUR NOTICE TO BUILDING MANAGEMENT FOR THIS WORK.

32. MECHANICAL CONTRACTOR TO BALANCE SCOPE OF WORK TO NEW AIR AND WATER QUANTITIES INDICATED UPON PROJECT COMPLETION. BALANCING TO BE PERFORMED BY BASE BUILDING APPROVED CONTRACTOR.

33. CONTRACTOR TO PROVIDE TRAVERSE VOLUMETRIC, STATIC PRESSURE AND TEMPERATURE READINGS AT BASE BUILDING MAIN SUPPLY, RETURN AND EXHAUST AIRWAYS. ALL DATA TO BE INCLUDED IN FINAL BALANCING REPORT.

34. REFER TO MECHANICAL PIPING PLAN FOR THERMOSTAT LOCATIONS AND THERMOSTAT CONTROL PLAN.

4	08/10/2026	ISSUE FOR BULLETIN #3
3	07/01/2026	ISSUE FOR ADD ALTERNATE
	05/22/2026	ISSUE FOR FILING
	04/03/2026	ISSUE FOR PRICING
NO.	DATE	ISSUE
REVISIONS		

PROJECT:

SNOWFLAKE

7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:

LEVEL 29 - MECHANICAL DUCTWORK PLAN

SEAL & SIGNATURE

DATE: 05/22/2026

SCALE: As indicated

PROJECT NO. _____

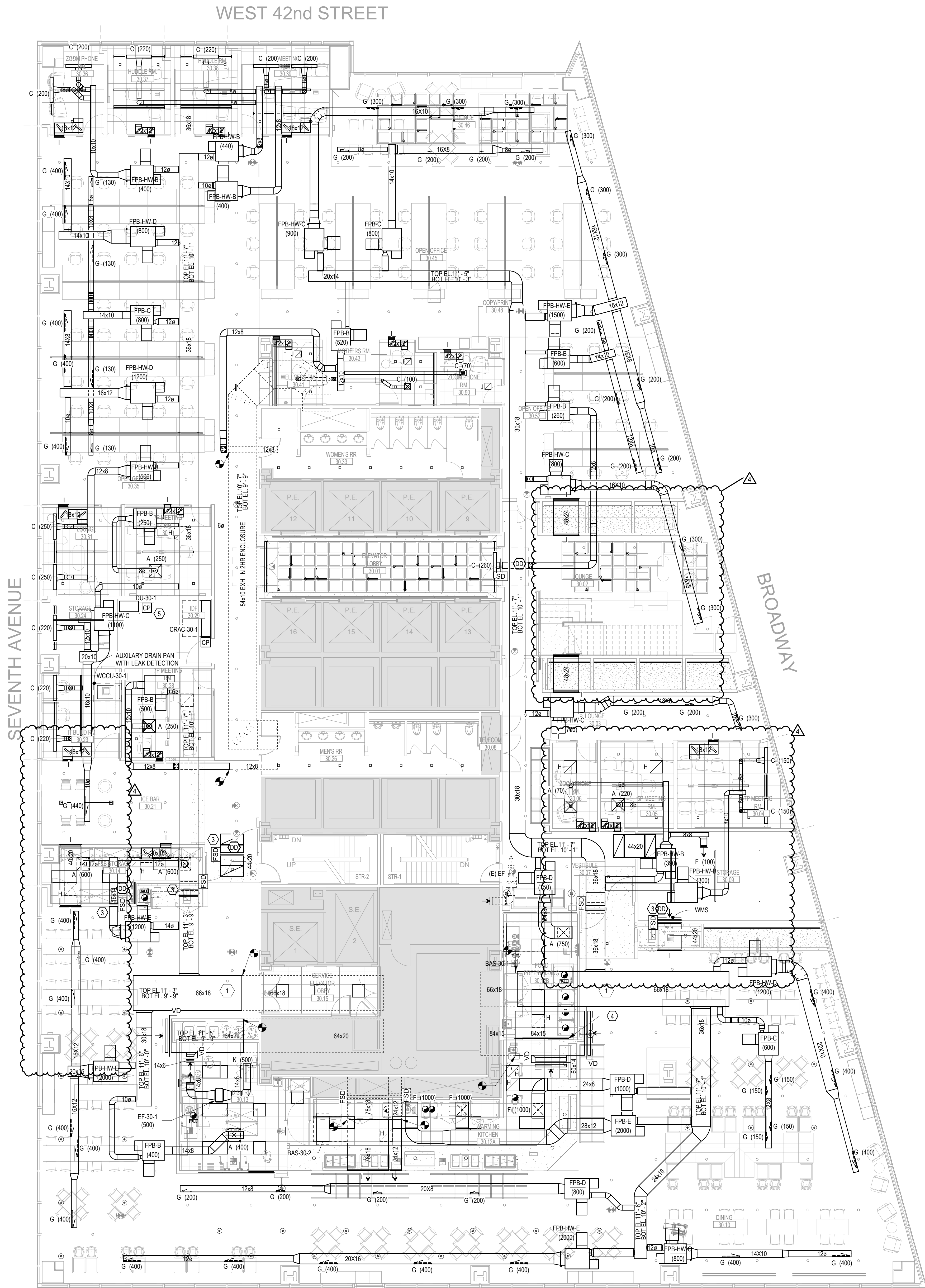
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NYC DOB NUMBER

M01397257-S1



KEY NOTES

- DUCTWORK TO BE CONSTRUCTED OF 16-GAUGE OUTER DUCT WITH ACOUSTICAL LINING, REGULAR LINER AND STANDARD GAUGE PERFORATED INNER DUCT TO MATCH BUILDING STANDARD FOR FAN DISCHARGE DUCTWORK.
- EXPOSED DUCTWORK SHALL BE OVAL OR ROUND SPIRAL DUCT CONSTRUCTED OF GALVANIZED STEEL WITH ACOUSTICAL LINING AS PER SPECIFICATIONS. FABRICATE AND INSTALL AS PER SMACNA STANDARDS, WITH ALL SEAMS AIRTIGHT AND CONNECTIONS SUITABLE FOR PRESSURE CLASS, SUPPORT AND HANG DUCT PER CODE REQUIREMENTS.
- PROVIDE PNEUMATIC PIPING AND WIRING AS REQUIRED FOR FIRE SMOKE DAMPER PER BUILDING RULES AND REGULATIONS. COORDINATE WITH BMS VENDOR.
- PROVIDE RETURN AIR TRANSFER OPENING(S) ABOVE THE CEILING WITH MINIMUM OF 1.90 SQ FEET FREE AREA AT PARTITION INDICATED. GC TO COORDINATE.
- UNIT LOCATION TO BE COORDINATED WITH IDF LAYOUT. LOCATION AND POSITIONING SHOULD ALLOW FOR ALL CLEARANCES AS SPECIFIED BY MANUFACTURER. IF CEILING HUNG, UNIT SHOULD BE VIBRATION INSULATED AS DESCRIBED FOR MECHANICAL EQUIPMENT IN SPECIFICATIONS.

MECHANICAL NOTES

- ALL WORK SHALL CONFORM TO THE LATEST BUILDING STANDARDS. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO OBTAIN A COPY OF THE BUILDING STANDARDS AND MEET WITH BUILDING MANAGEMENT IN ORDER TO BECOME TOTALLY FAMILIAR WITH THE BUILDING CONSTRUCTION RULES. THERE SHALL BE NO DEVIATION FROM THE BUILDING STANDARDS WITHOUT PRIOR WRITTEN APPROVAL FROM THE BUILDING MANAGEMENT.
- WHEN MECHANICAL WORK IS SUBCONTRACTED IT SHALL BE THE MECHANICAL CONTRACTORS RESPONSIBILITY TO COORDINATE SUBCONTRACTORS AND THE ASSOCIATED CONTRACTS. WHEN DISCREPANCIES ARISE PERTAINING TO WHICH CONTRACTOR PROVIDES A PARTICULAR ITEM OF THE MECHANICAL CONTRACT OR WHICH CONTRACTOR PROVIDES FINAL CONNECTIONS FOR A PARTICULAR ITEM OF THE MECHANICAL CONTRACT, IT SHALL BE BROUGHT TO THE ATTENTION OF THE MECHANICAL CONTRACTOR, WHOSE DECISION SHALL BE FINAL.
- ALL MISCELLANEOUS STEEL REQUIRED TO ENSURE PROPER INSTALLATION AND AS SHOWN IN DETAILS FOR PIPING, DUCTWORK, AND EQUIPMENT SHALL BE FURNISHED AND INSTALLED BY THE MECHANICAL CONTRACTOR AND COORDINATED WITH THE GENERAL CONTRACTOR FOR COORDINATION AMONG ALL TRADES.
- PROVIDE ALL MATERIALS, EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.
- INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURERS' RECOMMENDATIONS, CONTRACT DOCUMENTS, AND APPLICABLE CODES AND REGULATIONS.
- PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS CONNECTED TO AND WITHIN 50 FEET OR 50 PIPE DIAMETERS IF ISOLATED EQUIPMENT REFER TO SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL PIPING, DUCTWORK AND EQUIPMENT VIBRATION ISOLATION.
- LOCATE ALL TEMPERATURE, PRESSURE AND FLOW MEASURING DEVICES IN ACCESSIBLE LOCATIONS WITH STRAIGHT SECTION OF PIPE OR DUCT UP- AND DOWNSTREAM AS RECOMMENDED BY THE MANUFACTURER.
- COORDINATE ALL EQUIPMENT CONNECTIONS WITH MANUFACTURER'S CERTIFIED DRAWINGS. COORDINATE AND PROVIDE ALL DUCT AND PIPING TRANSITIONS REQUIRED FOR THE FINAL EQUIPMENT CONNECTIONS TO FURNISHED EQUIPMENT. FIELD VERIFY COORDINATE ALL DUCT AND PIPING DIMENSIONS BEFORE FABRICATIONS.
- THE LOCATIONS OF ALL ITEMS SHOWN ON THE DRAWINGS OR CALLED FOR IN THE SPECIFICATIONS THAT ARE NOT DEFINITELY FIXED BY DIMENSIONS ARE APPROXIMATELY ONLY. THE EXACT LOCATIONS NECESSARY TO SECURE THE BEST CONDITIONS AND RESULTS MUST BE DETERMINED BY THE PROJECT SITE CONDITIONS AND SHALL HAVE THE APPROVAL OF THE ENGINEER BEFORE BEING INSTALLED. DO NOT SCALE DRAWINGS.
- PROVIDE ACCESS PANELS FOR INSTALLATION IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVICE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER CONCEALED MECHANICAL EQUIPMENT. ACCESS PANELS SHALL BE TURNED OVER TO GENERAL CONTRACTOR INSTALLATION.
- ALL DUCTWORK, PIPING AND EQUIPMENT SUPPORTED FORM STRUCTURAL STEEL SHALL BE COORDINATED WITH GENERAL CONTRACTOR. ALL ATTACHMENTS TO STEEL BAR JOISTS, TRUSSES, OR JOIST GIRDERS SHALL BE AT PANEL POINTS. PROVIDE BEAM CLAMPS MEETING SMACNA STANDARDS.
- ALL OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUIT, ETC. SHALL BE FIRE STOPPED WITH FIRE STOPPING SIMILAR TO 3M OR APPROVED EQUAL AS SPECIFIED IN THE ARCHITECTURAL DOCUMENTS.
- ALL AIR CONDITIONING CONDENSATE DRAIN LINES FORM EACH AIR HANDLING UNIT AND ROOFTOP UNIT SHALL BE PIPED FULL SIZE OF THE UNIT. DRAIN OUTLET, WITH "P" TRAP, AND PIPED TO NEAREST DRAIN. SEE DETAILS SHOWN ON THE DRAWINGS OR THE CONTRACT SPECIFICATIONS FOR DEPTH OF AIR CONDITIONING CONDENSATE TRAP.
- CERTAIN ITEMS SUCH AS RISERS AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DAMPERS, ETC., ARE INDICATED ON THE CONTRACT DOCUMENT DRAWINGS FOR CLARITY FOR A SPECIFIC LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS. LOCATIONS OF ALL SUCH ITEMS SHALL BE INDICATED ON SHOP DRAWINGS BY THE INSTALLING CONTRACTOR FOR REVIEW AND APPROVAL DURING THE SHOP DRAWING PROCESS.
- EXACT LOCATIONS OF ALL THERMOSTATS SHALL BE AS PER ARCHITECT/ENGINEER. REFER TO ARCHITECTURAL DWGS FOR THERMOSTAT MOUNTING HEIGHT. MECHANICAL CONTRACTOR SHALL SUBMIT THERMOSTAT LOCATIONS FOR REVIEW PRIOR TO INSTALLATION.
- ALL DUCTWORK DIMENSIONS, AS SHOWN ON THE DRAWINGS, ARE INTERIOR DIMENSIONS AND DO NOT INCLUDE INCREASED DIMENSION TO ACCOUNT FOR DUCT LINING THICKNESS. REFER TO MECHANICAL SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL DUCT LINING.
- COORDINATE DIFFUSER, REGISTER, AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS, LIGHTING, AND OTHER CEILING ITEMS.
- LOCATE ALL MECHANICAL EQUIPMENT FOR UNOBSTRUCTED ACCESS TO UNIT ACCESS PANELS, CONTROLS AND VALVING.
- PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK SYSTEMS CONNECTED TO AIR HANDLING UNITS. FANS AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION, FLEXIBLE CONNECTIONS SHALL BE PROVIDED AT THE POINT OF CONNECTION TO THE EQUIPMENT UNLESS OTHERWISE INDICATED.
- ALL EQUIPMENT, DUCTWORK & PIPING SHALL BE INSTALLED AT MAXIMUM HEIGHTS ALLOWABLE OR AS REQUIRED TO CLEAR NEW CEILING. DUCTWORK & PIPING SHALL BE OFFSET AS REQUIRED TO AVOID CONFLICTS EVEN IF NOT SPECIFICALLY INDICATED ON PLAN.
- ALL EXPOSED SUPPLY DUCTWORK SHALL HAVE FLANGED AND GASKETTED CONNECTIONS AND SHALL NOT HAVE EXTERNAL INSULATION WRAP. PROVIDE ACOUSTIC LINING AS REQUIRED BY SPECIFICATIONS.
- ALL EXPOSED DUCTWORK SHALL BE FREE OF DEFORMITIES, WRITING, STICKERS OR ANY OTHER MARKINGS. DUCT SURFACES SHALL BE TREATED AS REQUIRED FOR PAINTING. REFER TO ARCHITECTURAL DWGS FOR LOCATIONS OF DUCTWORK TO BE PAINTED.
- ALL DUCTWORK EXPOSED TO MOISTURE SHALL BE ALUMINUM FOR A MIN. OF 10 FEET.
- ALL SUPPLY DUCTWORK SHOWN TO HAVE ROUND DIMENSIONS SHALL BE SPIRAL ROUND DUCTWORK.
- MOUNT ALL SUPPLY REGISTERS AT SAME ELEVATION. ELEVATIONS SHALL BE INDICATED ON THE SHEET/ETAL SHOP DWGS FOR REVIEW BY ARCHITECT/ENGINEER.
- FOR ROOMS WITH EQUIPMENT RACKS, PRODUCTION OF SHEETMETAL SHOP DRAWINGS SHALL INCLUDE ALL COORDINATION REQUIRED TO INDICATE EXACT LOCATIONS OF ALL TECHNICAL EQUIPMENT, EQUIPMENT & LADDER RACK SYSTEMS, ETC. SO AS TO ACCURATELY REVIEW COORDINATED PLACEMENT OF ALL AIR OUTLETS SERVING CRITICAL AREAS.
- PROVIDE ACCESS DOORS IN DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, AND OTHER ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.
- SMOKE DETECTORS SHALL BE FURNISHED AND WIRED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR MOUNTING THE SMOKE DETECTOR IN DUCTWORK AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS.
- MECHANICAL CONTRACTOR TO TAKE PRE-CONSTRUCTION AIR READINGS AT ALL BASE BUILDING MAIN DUCTWORK SERVING SCOPE OF WORK AREA. INCLUDE KEYED PLAN SHOWING READING LOCATIONS AND SUBMIT TO ENGINEER PRIOR TO PRODUCTION OF SHEET METAL SHOP DWGS. REFER TO SPECIFICATIONS FOR MORE INFORMATION.
- AIR BALANCING WORK SHALL BE WITNESSED BY THE BUILDING'S CHIEF ENGINEER OR HIS REPRESENTATIVE AND A COPY OF THE CERTIFIED BALANCING REPORT SHALL BE SUBMITTED TO THE BUILDING MANAGEMENT. PROVIDE 48 HOUR NOTICE TO BUILDING MANAGEMENT FOR THIS WORK.
- MECHANICAL CONTRACTOR TO BALANCE SCOPE OF WORK TO NEW AIR AND WATER QUANTITIES INDICATED UPON PROJECT COMPLETION. BALANCING TO BE PERFORMED BY BASE BUILDING APPROVED CONTRACTOR.
- CONTRACTOR TO PROVIDE TRAVERSE VOLUMETRIC, STATIC PRESSURE AND TEMPERATURE READINGS AT ALL BASE BUILDING MAIN SUPPLY, RETURN AND EXHAUST SHAFTS. READINGS TO BE INCLUDED IN FINAL BALANCING REPORT.
- REFER TO MECHANICAL PIPING PLAN FOR THERMOSTAT LOCATIONS AND THERMOSTAT CONTROL PLAN.

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4	08/10/2026	ISSUE FOR BULLETIN #3
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	04/03/2026	ISSUE FOR PRICING

NO.	DATE	REVISIONS	ISSUE
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PROJECT:

SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:

LEVEL 30 - MECHANICAL DUCTWORK PLAN

SEAL & SIGNATURE

DATE: 05.22.2026

SCALE: As indicated

PROJECT NO. ---

DRAWN: CHECKED:

DRAWING NO.

M-530.00

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NYC DOB NUMBER

M01397257-S1

[illegible]

- 1 DUCTWORK TO BE CONSTRUCTED OF 16-GAUGE OUTER DUCT WITH "ACOUSTICAL LINING, TIED LINER AND STANDARD GAUGE PERFORATED INNER DUCT TO MATCH BUILDING STANDARD FOR FAN DISCHARGE DUCTWORK.
- 2 EXPOSED DUCTWORK SHALL BE OVAL OR ROUND SPIRAL DUCT CONSTRUCTED OF GALVANIZED STEEL WITH ACOUSTICAL LINING AS PER SPECIFICATIONS, FABRICATE AND INSTALL AS PER SMACNA STANDARDS, WITH ALL SEAMS, TIGHTEN AND CONNECTIONS SUITABLE FOR PRESSURE CLASS, SUPPORT AND HANG DUCT PER CODE REQUIREMENTS.
- 3 UNIT LOCATION TO BE COORDINATED WITH IDP LAYOUT, LOCATION AND POSITIONING SHALL BE SHOWN ON WORK ORDER VARIATIONS AS SPECIFIED BY MAIN CONTRACTOR. IF CEILING HUNG, UNIT SHOULD BE VIBRATION INSULATED AS DESCRIBED FOR MECHANICAL EQUIPMENT IN SPECIFICATIONS.

1. ALL WORK SHALL CONFORM TO THE LATEST BUILDING STANDARDS. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO OBTAIN A COPY OF THE BUILDING STANDARDS MEET THE CITY OF CHICAGO. THE CONTRACTOR SHALL BE TOTALLY FAMILIAR WITH THE BUILDING CONSTRUCTION RULES. THERE SHALL BE NO DEVIATION FROM THE BUILDING STANDARDS WITHOUT PRIOR WRITTEN APPROVAL FROM THE BUILDING DEPARTMENT.

2. WHEN MECHANICAL WORK IS SUBCONTRACTED IT SHALL BE THE MECHANICAL CONTRACTORS RESPONSIBILITY TO COORDINATE SUBCONTRACTORS AND THE ASSOCIATED CONTRACTS. WHEN DISCREPANCIES ARISE PERTAINING TO WHICH CONTRACTOR PROVIDES A PARTICULAR ITEM OF THE MECHANICAL CONTRACT OR WHEN DISCREPANCIES PERTAIN TO THE MECHANICAL CONTRACTORS PART OF THE MECHANICAL CONTRACT, IT SHALL BE BROUGHT TO THE ATTENTION OF THE MECHANICAL CONTRACTOR, WHOSE DECISION SHALL BE FINAL.

3. ALL MISCELLANEOUS STEEL REQUIRED TO ENSURE PROPER INSTALLATION AND AS SHOWN IN DETAILS FOR PIPING, DUCTWORK, AND EQUIPMENT SHALL BE FURNISHED AND INSTALLED BY THE CONTRACTOR. THE CONTRACTOR SHALL COORDINATE WITH THE GENERAL CONTRACTOR FOR COORDINATION AMONG ALL TRADES.

4. PROVIDE ALL MATERIALS, EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

5. INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS, CONTRACT DOCUMENTS, AND APPLICABLE CODES AND REGULATIONS.

6. PROVIDE VIBRATION ISOLATION FOR ALL MECHANICAL EQUIPMENT TO PREVENT TRANSMISSION OF VIBRATION TO BUILDING STRUCTURE.

7. WITHIN 50 FEET OR 50 PIPE DIAMETERS IF ISOLATED EQUIPMENT REFER TO SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL PIPING, DUCTWORK AND EQUIPMENT INSTALLATION.

8. LOCATE ALL TEMPERATURE, PRESSURE AND FLOW MEASURING DEVICES IN ACCORDANCE WITH THE DESIGN OR WEIGHT OF THE PIPING DUCT UP- AND DOWNSTREAM AS RECOMMENDED BY THE MANUFACTURER.

9. COORDINATE ALL EQUIPMENT CONNECTIONS WITH MANUFACTURER'S CERTIFIED DRAWINGS. COORDINATE AND PROVIDE ALL DUCT AND PIPING TRANSITIONS REQUIRED FOR THE FINAL EQUIPMENT CONNECTIONS TO FURNISHED EQUIPMENT. FIELD VERIFY COORDINATE ALL DUCT AND PIPING DIMENSIONS BEFORE FABRICATIONS.

10. THE LOCATIONS OF ALL ITEMS SHOWN ON THE DRAWINGS OR CALLED FOR IN THE SPECIFICATIONS SHALL BE EXACTLY LOCATED. THE CONTRACTOR SHALL BE RESPONSIBLE FOR APPROXIMATELY 10% THE EXACT LOCATIONS NECESSARY TO SECURE THE BEST CONDITIONS AND RESULTS MUST BE DETERMINED BY THE PROJECT SITE CONDITIONS AND THE APPROVAL OF THE ENGINEER BEFORE BEING INSTALLED. DO NOT SCALE DRAWINGS.

11. PROVIDE ACCESS PANELS FOR INSTALLATION IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVICE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER CONCEALED MECHANICAL EQUIPMENT. ACCESS PANELS SHALL BE TURNED OVER TO GENERAL CONTRACTOR INSTALLATION.

12. ALL DUCTWORK, PIPING AND EQUIPMENT SUPPORTED FROM STRUCTURAL STEEL SHALL BE COORDINATED WITH GENERAL CONTRACTOR. ALL ATTACHMENTS TO STEEL BAR JOISTS, TRUSSES, OR JOIST GIRDERS SHALL BE AT PANEL POINTS. PROVIDE BEAM CLAMPS MEETING SAME STANDARDS.

13. ALL OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUIT, ETC. SHALL BE FIRE STOPPED WITH A PRODUCT SIMILAR TO 3M OR APPROX EQUIVAL AS SPECIFIED IN MECHANICAL CONTRACT DOCUMENTS.

14. ALL AIR CONDITIONING CONDENSATE DRAIN LINES FORM EACH AIR HANDLING UNIT AND ROOFTOP UNIT SHALL BE PIPED FULL SIZE OF THE UNIT DRAIN OUTLET, WITH "P" TRAP, AND PIPED TO NEAREST DRAIN. SEE DETAILS SHOWN ON THE DRAWINGS OR THE CONTRACT SPECIFICATIONS FOR DEPTH OF AIR CONDITIONING CONDENSATE TRAP.

15. CERTAIN ITEMS SUCH AS RISERS AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DAMPERS, ETC., ARE INDICATED ON THE CONTRACT DOCUMENT DRAWINGS FOR THE CONTRACTOR TO PROVIDE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE DESIGN AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS. LOCATIONS OF ALL SUCH ITEMS SHALL BE INDICATED ON SHOP DRAWINGS BY THE INSTALLING CONTRACTOR, FOR REVIEW AND APPROVAL OF THE ENGINEER BEFORE SHOP DRAWINGS ARE SUBMITTED.

16. EXACT LOCATIONS OF ALL THERMOSTATS SHALL BE AS PER ARCHITECT/ENGINEER. REFER TO ARCHITECTURAL DWGS FOR THERMOSTAT MOUNTING HEIGHT. MECHANICAL CONTRACTOR SHALL SUBMIT THERMOSTAT LOCATIONS FOR REVIEW PRIOR TO INSTALLATION.

17. ALL DUCTWORK DIMENSIONS, AS SHOWN ON THE DRAWINGS, ARE INTERIOR DIMENSIONS. DO NOT ADD TO THE DIMENSIONS. THE CONTRACTOR SHALL PROVIDE DUCT LINING THICKNESS. REFER TO MECHANICAL SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL DUCT LINING.

18. COORDINATE DIFFUSER, REGISTER, AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS, LIGHTING, AND OTHER CEILING ITEMS.

19. LOCATE ALL MECHANICAL EQUIPMENT TO UNOBSTRUCTED ACCESS TO UNIT ACCESS PANELS, CONTROLS AND VALVING.

20. PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK SYSTEMS CONNECTED TO HAND VULCANIZING, FANS AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION. FLEXIBLE CONNECTIONS SHALL BE INSTALLED AT THE POINT OF CONNECTION TO THE EQUIPMENT UNLESS OTHERWISE INDICATED.

21. ALL EQUIPMENT, DUCTWORK & PIPING SHALL BE INSTALLED AT MAXIMUM HEIGHTS ALLOWABLE OR AS REQUIRED TO CLEAR NEW CEILING. DUCTWORK & PIPING SHALL BE OFFSET AS REQUIRED TO AVOID CONFLICTS EVEN IF NOT SPECIFICALLY INDICATED ON PLAN.

22. ALL EXPOSED SUPPLY DUCTWORK SHALL HAVE FLANGED AND GASKETTED CONNECTIONS. PROVIDE SOUND ATTENUATION EQUIPMENT AS REQUIRED. PROVIDE ACOUSTIC LINING AS REQUIRED BY SPECIFICATIONS.

23. ALL EXPOSED DUCTWORK SHALL BE FREE OF DEFORMITIES, WRITING, STICKERS OR ANY OTHER MARKINGS. DUCT SURFACES SHALL BE TREATED AS REQUIRED FOR PAINTING. REFER TO ARCHITECTURAL DWGS FOR LOCATIONS OF DUCTWORK TO BE PAINTED.

24. ALL DUCTWORK EXPOSED TO MOISTURE SHALL BE ALUMINUM FOR A MIN. OF 10 FEET.

25. ALL SUPPLY DUCTWORK SHALL HAVE ROUND DIMENSIONS SHALL BE SPIRAL ROUND DUCTWORK.

26. MOUNT ALL SUPPLY REGISTERS AT SAME ELEVATION. ELEVATIONS SHALL BE INDICATED ON THE SHEET/META. SHOP DWGS FOR REVIEW BY ARCHITECT/ENGINEER.

27. FOR ROOMS WITH EQUIPMENT RACKS: PRODUCTION OF SHEETMETAL SHOP DRAWINGS SHALL INCLUDE ALL COORDINATION REQUIRED TO LOCATE EXACT EQUIPMENT RACKS. THE CONTRACTOR SHALL PROVIDE EQUIPMENT RACKS. ITEMS, ETC. AS SO AS ACCURATELY REVER COORDINATED PLACEMENT OF ALL AIR OUTLETS SERVING CRITICAL AREAS.

28. PROVIDE ACCESS DOORS TO DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, AND OTHER ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.

29. SMOKE DETECTORS SHALL BE FURNISHED AND WIRED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR ROUTING SMOKE DETECTORS THROUGH THE DUCTWORK AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS.

30. MECHANICAL CONTRACTOR TO TAKE PRE-CONSTRUCTION AIR READINGS AT ALL BASE BUILDING MAIN DUCTWORK SERVING SPACE OF WORK AREA. INCLUDE KEYPAD PLAN SHOWING READING LOCATIONS AND SUBMIT TO ENGINEER PRIOR TO PRODUCTION OF SHEET METAL SHOP DWGS. REFER TO SPECIFICATIONS FOR MORE INFORMATION.

31. AIR BALANCING WORK SHALL BE WITNESSED BY THE BUILDING'S CHIEF ENGINEER OR A REGISTERED MECHANICAL ENGINEER. THE CONTRACTOR SHALL SUBMIT THE RESULTS SUBMITTED TO THE BUILDING MANAGEMENT. PROVIDE 48 HOUR NOTICE TO BUILDING MANAGEMENT FOR THIS WORK.

32. MECHANICAL CONTRACTOR TO BALANCE SCOPE OF WORK TO NEW AIR AND WATER QUANTITIES INDICATED UPON PROJECT COMPLETION. BALANCING TO BE PERFORMED BY BASE BUILDING APPROVED CONTRACTOR.

33. CONTRACTOR TO PROVIDE TRAVERSE VOLUMETRIC, STATIC PRESSURE AND AIR FLOW MEASURING EQUIPMENT. CONTRACTOR SHALL PROVIDE AIR AND EXHAUST SHAFTS. READINGS TO BE INCLUDED IN FINAL BALANCING REPORT.

34. REFER TO MECHANICAL PIPING PLAN FOR THERMOSTAT LOCATIONS AND THERMOSTAT CONTROL PLAN.

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4	08/10/2026	ISSUE FOR BULLETIN #3
3	07/01/2026	ISSUE FOR ADD ALTERNATE
	05/22/2026	ISSUE FOR FILING
1	04/22/2026	ADDENDUM #1
	04/03/2026	ISSUE FOR PRICING
NO.	DATE	ISSUE
REVISIONS		

SNOWFLAKE

7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

LEVEL 31 - MECHANICAL DUCTWORK PLAN

SEAL & SIGNATURE	DATE:		05.22.2026
	SCALE:		As indicated
	PROJECT NO.		---
	DRAWN:	CHECKED:	
	DRAWING NO.		
M-531.00			

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